

WPA-ing Out

 **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

I thought that my password was super-secret, but it turns out that passwords passed over the AIR can be CRACKED, especially if I used the same wireless network password as one in the rockyou.txt credential dump.

Use this pcap file and the rockyou wordlist. The flag should be entered in the picoCTF{XXXXXX} format.

Hints

- Finding the IEEE 802.11 wireless protocol used in the wireless traffic packet capture is easier with Wireshark, the JAWS of the network.
 - Aircrack-ng can make a pcap file catch big air...and crack a password.
-

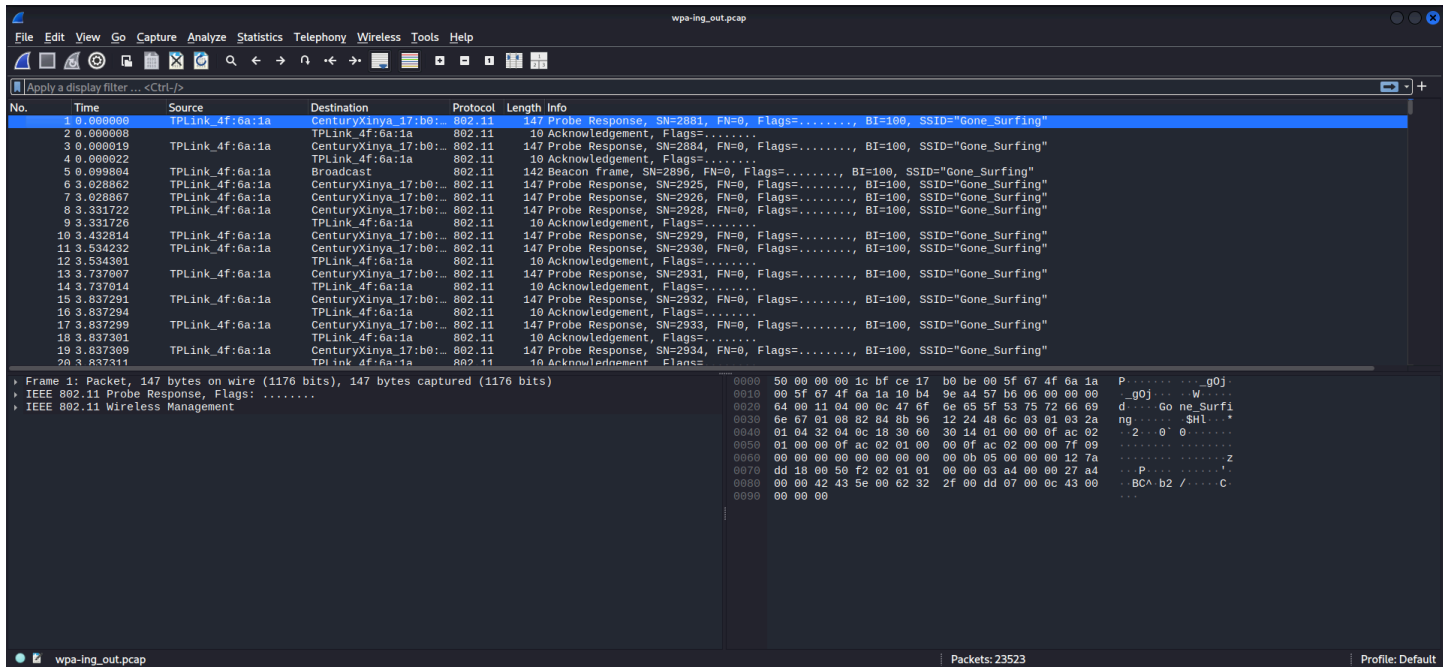
Tools Used

- `file` – Determine file type
 - **Wireshark** – GUI tool for analyzing wireless (802.11) network traffic
 - `aircrack-ng` – Wireless network password cracking tool
 - `/usr/share/wordlists/rockyou.txt` – Common password wordlist
-

Complete Solution

Step 1: Download and Prepare

Download the PCAP file named `wpa-ing_out.pcap` from the challenge page.



Step 2: Basic File Inspection

First, verify the file type and general information:

```
file wpa-ing_out.pcap
```

Expected output:

```
wpa-ing_out.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4
(802.11, capture length 65535)
```

The file is a network packet capture containing **IEEE 802.11 wireless traffic** (Wi-Fi).

Step 3: Understand the Challenge

This is a **WPA/WPA2 password cracking** challenge. The PCAP file contains a **4-way handshake** between a client and a wireless access point. When a device connects to a WPA/WPA2 protected Wi-Fi network, it performs a 4-way handshake that can be captured and used to crack the password offline.

How it works:

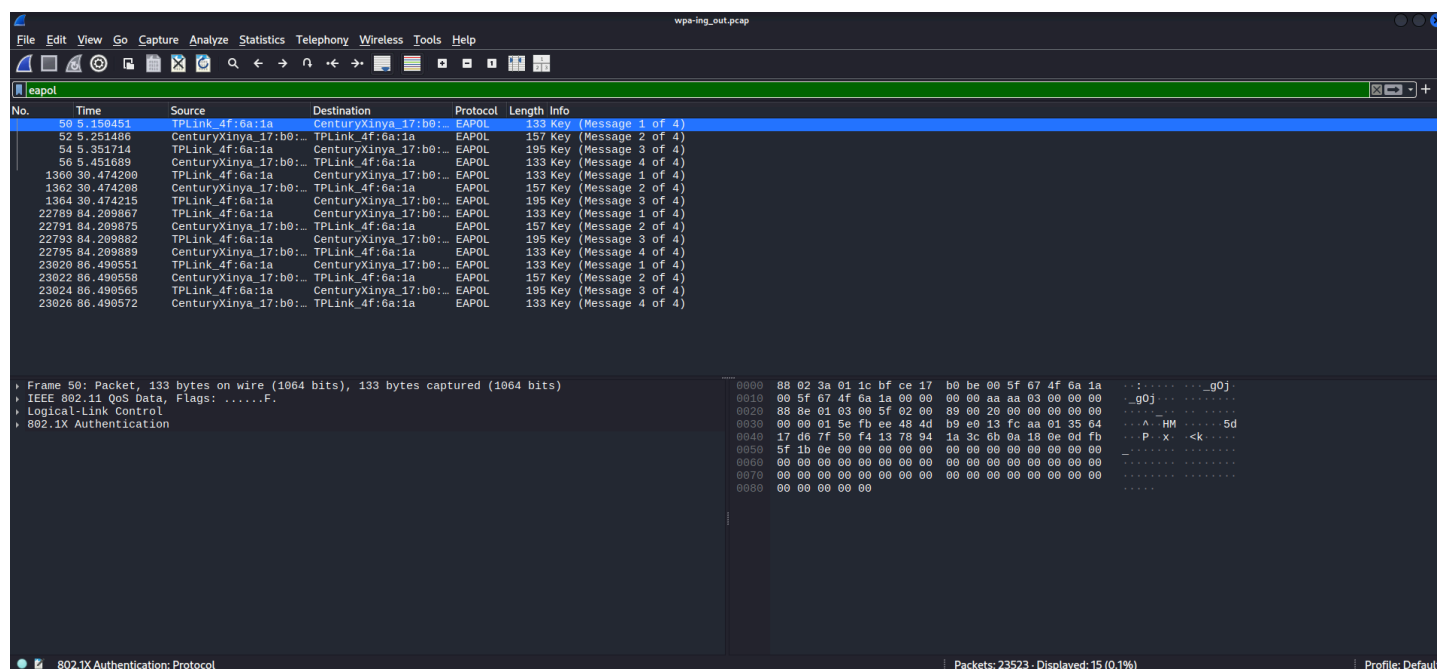
1. The handshake contains encrypted information derived from the password

2. A tool like `aircrack-ng` can try passwords from a wordlist (`rockyou.txt`) against the handshake
3. When a password correctly decrypts the handshake, it is the network password

Step 4: Verify the Handshake in Wireshark (Optional)

Open the PCAP in Wireshark to verify the presence of EAPOL packets (which indicate the 4-way handshake):

1. Open `wpa-ing_out.pcap` in Wireshark
2. Apply the display filter: `eapol`



This confirms the capture contains a valid WPA handshake that can be cracked.

Step 5: Crack the Password with Aircrack-ng

Use `aircrack-ng` with the `rockyou.txt` wordlist to crack the WPA password:

```
aircrack-ng -w /usr/share/wordlists/rockyou.txt wpa-ing_out.pcap
```

Expected output (abbreviated):

Aircrack-ng 1.7

[00:00:00] 1138/10303727 keys tested (3124.66 k/s)

Time left: 54 minutes, 57 seconds

0.01%

KEY FOUND! [<REDACTED>]

Master Key : <REDACTED>

Transient Key : <REDACTED>

EAPOL HMAC : <REDACTED>

Key finding: The cracked password is <REDACTED>

Step 6: Format the Flag

The challenge requires the flag in the format `picoCTF{XXXXXX}` . Simply wrap the cracked password in the flag format:

```
picoCTF{<REDACTED>}
```

Final Result

```
picoCTF{<REDACTED>}
```

Note: The actual flag is redacted here. In the real challenge, the extracted file will contain the complete flag.

Key Concepts

Concept	Description
IEEE 802.11	The standard for wireless local area networks (Wi-Fi).
WPA/WPA2	Security protocols for Wi-Fi networks that use a pre-shared key (password).
4-Way Handshake	The authentication process between a client and access point; capturing it allows offline password cracking.
EAPOL (Extensible Authentication Protocol over LAN)	The protocol used during the 4-way handshake.
Aircrack-ng	A suite of tools for auditing wireless networks, including WPA/WPA2 password cracking.
rockyou.txt	A common wordlist containing millions of real-world passwords (from a 2009 data breach).
Offline Cracking	Capturing encrypted data and attempting passwords locally without interacting with the target.

Pro Tips

1. **Always verify the handshake** – Before running `aircrack-ng`, open the PCAP in Wireshark and filter for `eapol`. If you don't see at least 4 packets, the capture may not contain a complete handshake.
2. **Use the correct wordlist** – The hint specifically mentions `rockyou.txt`. It's located at `/usr/share/wordlists/rockyou.txt` on Kali Linux. You may need to unzip it first: `sudo gunzip /usr/share/wordlists/rockyou.txt.gz`
3. **Be patient** – Cracking can take time depending on the wordlist size and password complexity.
4. **Check the BSSID** – `aircrack-ng` will show the network's BSSID (MAC address) and ESSID (network name). Make sure you're targeting the correct network.
5. **EAPOL packets are key** – These are the packets that contain the handshake data needed for cracking.